Run report: invariant against raw guesses at $\ell = 8$

- **Study:** `research/dreg_invariant_guess_20261029/`, pre-registered (commit `d0b05eee0`). Amendment 1, made after the data, replaces the soundness rule and is disclosed as post hoc.
- **Run:** 2026-10-29, 03:08 to 11:53 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.

What ran

- **Tool:** `examples/dreg_linear_guess.rs` specialises the m = 3 chained S_3 system. `--fix v=b` fixes a variable; `--sum j=i+c` substitutes `x_j := x_i + c`. Then it scans the MutantXL closure (dense, at most 140,000 columns) at degrees 3 and 4.
- **Exactness:** `--check` evaluates every specialised equation against the original at 4,096 random points satisfying the guesses. It is true on every scored run, and a deliberately wrong build fails it.
- **Draws:** the committed (22, 8) draws (N = 46). Unsatisfiable draws 0, 3, 5, 6; satisfiable draw 1.
- **Arms:** three guesses each, 8 branches, 43 occurring variables, 136,698 degree-4 columns.
  - S3: $x_{15} := x_7 + c$ (the top coordinate of $s = x_1 + x_2$), x_{23}, x_{24}.
  - X3: x_7, x_{23}, x_{24}.
- **Early stop:** an (arm, draw) stops at its first branch that saturates degree 4 without refuting.
- **Limits:** `prlimit --as=15e9`, `timeout 7200`, one process at a time, on the 4-core, 16 GB container, not isolated. Iso Lab EKS is not reachable from it.

Completion

- 42 processes exited 0, and 5 were killed with exit 137:
  - S3 draw 0 branch 3, twice: an unsatisfiable draw, unmeasured;
  - S3 branches 1, 5 and 6 on satisfiable draw 1, which are the branches holding its solutions.
- Two container restarts each killed one run in flight. Their empty outputs are in `runs/killed-by-restart/`.
- Peak RSS of the completed runs was 6.1–8.5 GB.

Measured values

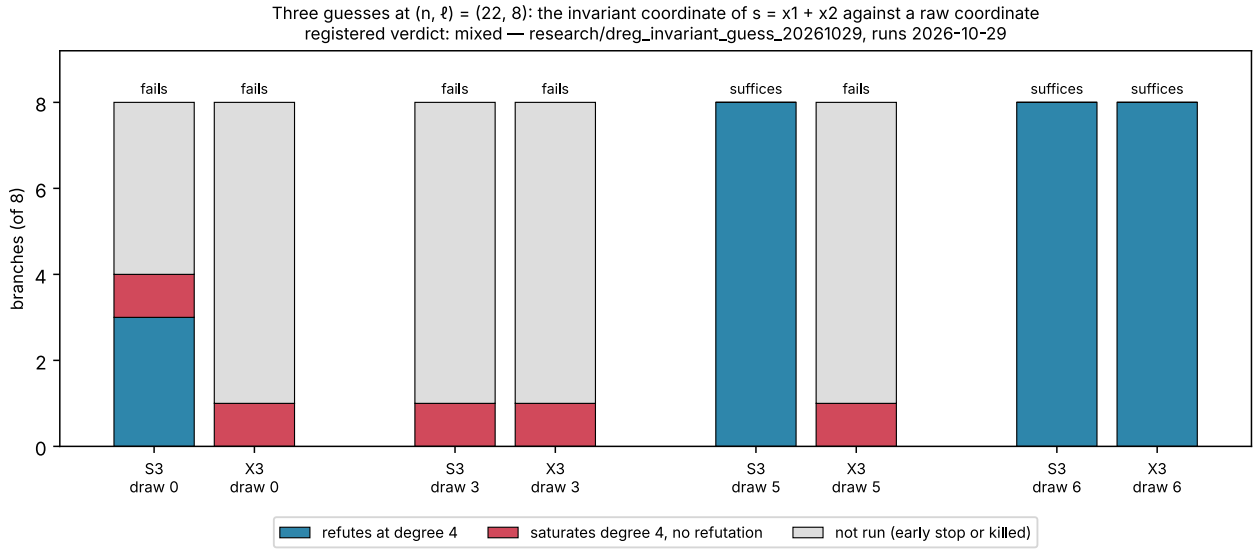


Figure 1: Branches per arm and draw: refuting at degree 4, saturating degree 4 without refutation, or not run. Source: `plot.py` (layout only) on `plot-data.json` from `dreg_linear_guess_score --plot-data`.

- **Controls:**
  - the instrument identity control matches the committed G3 record on every non-timing field;
  - the positive control refutes on 4 of 4;
  - soundness (V3′, post hoc): the 6 solutions of draw 1 lie only in the killed branches, and the 5 refuting branches hold none.
  - Valid.
- **S3:** suffices on draws 5 and 6, and fails on 0 and 3.
- **X3:** suffices on draw 6 only.

- **Registered verdict:** mixed (S3 split, X3 fails).
- **Per-run values:** in `score-output.txt` and `runs/`.